claude-windows / a9fd4426-f234-4e4a-bfb4-1f2c8255c661

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-muneem\a9fd4426-f234-4e4a-bfb4-1f2c8255c661\subagents\workflows\wf_076beba8-59c\agent-ad39fa3d46ebbb04d.jsonl`
- SHA-256: `37f374a02dfbea0457dcdc64a83987aed68c79196b47cf05e8406dd92362e372`
- Source modified: `2026-05-31T04:26:38+00:00`
- Imported at: `2026-07-05T16:48:21+00:00`
- project: `wf_076beba8-59c`
- session_id: `a9fd4426-f234-4e4a-bfb4-1f2c8255c661`

Transcript

1. user (2026-05-31T04:26:37.959Z)

Synthesis: research report

```
**Question:** Identify the best LOCAL (self-hosted, single NVIDIA CUDA GPU) vision / document-AI
/ OCR models ? and, as a clearly-flagged approval-gated fallback only, paid cloud API services ?
for parsing BANK STATEMENT PDFs into structured transaction tables (date, narration/description,
withdrawal/debit, deposit/credit, running balance), for a LOCAL-FIRST personal-finance tool
("muneem").

Target documents: Indian bank statements (HDFC, Axis, ICICI, AU Small Finance) ? dense, multi-
column, often SEMI-RULED or borderless tables, sometimes multiple accounts per statement; both
born-digital (text layer present) and the harder cases where pdfplumber's
extract_tables/coordinate clustering is unreliable (e.g. Axis, whose column layout varies
between statements).

HARD CONSTRAINTS (a model that violates these is unusable for us):
1. MUST run 100% LOCALLY for statement contents (highly sensitive). NO cloud APIs by default.
Paid/cloud is acceptable ONLY as an explicitly-approved fallback ? still report the best cloud
options but label them clearly as "cloud/sensitive ? approval-gated, not default".
2. MUST be usable from Python (transformers / vLLM / Ollama / ONNX Runtime / native lib).
3. License MUST be permissive & commercially safe for BOTH the CODE and the MODEL WEIGHTS: MIT,
BSD, Apache-2.0, ISC, HPND. AVOID and explicitly FLAG anything GPL/LGPL/AGPL, CC-BY-NC / non-
commercial, research-only, or custom "community"/acceptable-use licenses with restrictions. This
is critical ? many document-AI models and their training datasets are non-commercial (e.g.
LayoutLMv3 weights are CC-BY-NC; Surya and Marker use GPL-or-commercial dual licensing; several
VLM weights ship custom restrictive licenses). Verify the WEIGHTS license, not just the repo
license.
4. Hardware: a single consumer/prosumer NVIDIA GPU. Give VRAM tiers (~8-12 GB consumer, ~16-24
GB) and which models/quantizations fit each.

Evaluate and compare these LOCAL approaches ? for each give: code license, MODEL-WEIGHTS license
(state explicitly), latest release / maintenance status as of 2025-2026, VRAM need +
quantization options, Python integration path, and reliability specifically on DENSE FINANCIAL
TABLES / multi-column statements:
- Open vision-language models (VLMs) for document/table understanding: Qwen2-VL and Qwen2.5-VL
(note per-size license ? which sizes are Apache-2.0 vs restricted), MiniCPM-V 2.6, InternVL2 /
2.5, Microsoft Phi-3.5-vision, Mistral Pixtral, Meta Llama-3.2-Vision (flag the Llama Community
License restrictions), Moondream2, GOT-OCR2.0, olmOCR and other OCR-specialized VLMs.
- OCR-specialist / document-AI pipelines: PaddleOCR + PP-Structure (table recognition), docTR,
EasyOCR, Tesseract (already used), Microsoft Table Transformer / TATR (check code AND
PubTables-1M dataset/weights license), Donut, Surya (FLAG license), Marker (FLAG license),
MinerU / PDF-Extract-Kit, RapidOCR.
- Table-structure recognition specifically (cell/grid reconstruction for borderless tables):
TATR, PP-Structure, UniTable, etc.
```

Also assess: for SEMI-RULED / borderless bank tables, is a prompt-driven VLM (page image -> JSON transactions) more reliable than a classic OCR + table-structure pipeline? What are the accuracy/hallucination risks of VLMs on financial NUMBERS (transposed digits, dropped/duplicated rows), and how should output be VERIFIED? (muneem already machine-checks parses by balance reconciliation ? opening + sum(deposits) - sum(withdrawals) == closing, plus per-row running-balance walks and cross-checks against the bank's printed summary ? so any model's output can be automatically validated, which should inform the recommendation.)

Finally, briefly cover the best PAID/CLOUD fallbacks (clearly flagged approval-gated for sensitive data): AWS Textract (AnalyzeExpense / Tables), Google Document AI, Azure AI Document Intelligence, Reducto, LlamaParse, Anthropic/OpenAI vision ? note any self-hosting option and data-handling terms.

Deliver a RANKED shortlist of the most promising LOCAL options for muneem (NVIDIA GPU, Python, commercial-license-safe), each with license (code + weights), VRAM tier, maturity, strengths/weaknesses for dense bank-statement tables, and a concrete recommendation: which local model(s) to prototype for muneem's statement parsing and how they'd combine with the existing deterministic parsers + balance-reconciliation verification (e.g. VLM as a fallback/cross-check when deterministic parsing fails to reconcile). Cite sources and call out any license uncertainty explicitly.

21 claims survived 3-vote adversarial verification. Merge semantic duplicates and synthesize.

Confirmed claims

[0] olmOCR 2 releases its model weights (olmOCR-2-7B-1025), training data, and code under Apache 2.0, a permissive commercially-s

Vote: 3-0 · Source: https://arxiv.org/html/2510.19817v1 (primary)
Quote: "We release our model, data and code under permissive open licenses. [Table 1: olmOCR 2 ? Apache 2.0; Chandra OCR ? OpenRAIL-M]"
Verifier evidence (high): Core claim VERIFIED by primary sources, with one minor (non-material) inaccuracy.

CONFIRMED:
1. olmOCR-2-7B-1025 WEIGHTS = Apache 2.0. HF model card (https://huggingface.co/allenai/olmOCR-2-7B-1025) metadata header reads "license: apache-2.0"; model is fine-tuned from Qwen2.5-VL-7B-Instruct (also Apache 2.0). Arxiv 2510.19817v1 Table 1 lists olmOCR 2 = Apache 2.0.
2. CODE = Apache 2.0 (arxiv + olmOCR repo).
3. Chandra OCR weights = OpenRAIL-M, restrictive ? CONFIRMED and actually UNDERSTATED. GitHub datalab-to/chandra/MODEL_LICENSE header: "AI PUBS OPEN RAIL-M LICENSE (MODIFIED)" with $2M revenue/funding cap + competitive-use ban. This is exactly the kind of non-commercial/restricted weights license muneem's CLAUDE.md flags. (Chandra's Python code is Apache 2.0, but weights ? the load-bearing part ? are not.)

INACCURACY (not fatal): The claim says training DATA is "Apache 2.0." It is not ? olmOCR-mix-1025 dataset metadata (https://huggingface.co/datasets/allenai/olmOCR-mix-1025) reads "License: odc-by" (Open Data Commons Attribution). The arxiv Table 1 lumps "model, data and code" under one permissive-license checkmark, which the claim over-specified as "Apache 2.0." However, ODC-BY is itself permissive and commercially safe (attribution-only, no non-commercial clause), so the practical conclusion ? olmOCR 2 is license-safe for muneem, unlike Chandra ? is unaffected.

Both model cards also note "intended for research and educational use per Ai2 Responsible Use Guidelines," but the binding license tag is Apache 2.0 (an aspirational guideline, not a license restriction), so this does not negate commercial usability.

Verdict: central licensing claim (olmOCR2 weights+code Apache 2.0; Chandra restrictive) is well-supported by primary sources (HF metadata, arxiv, GitHub license file), current (Oct 2025 "1025" release), and not marketing. The "training data = Apache 2.0" detail is technically wrong (ODC-BY) but immaterial since ODC-BY is still permissive. NOT refuted.

[1] olmOCR 2 is built on Qwen2.5-VL-7B-Instruct, a 7B-parameter vision-language model, fine-tuned then RL-trained; this anchors its V

Vote: 3-0 · Source: https://arxiv.org/html/2510.19817v1 (primary)
Quote: "We start with a Qwen2.5-VL-7B-Instruct model that has been fine-tuned on olmOCR-mix-1025."

Verifier evidence (high): All four sub-claims are confirmed by PRIMARY sources. (1) Base model: arXiv 2510.19817v1 states "olmOCR 2 is powered by olmOCR-2-7B-1025... We start with a Qwen2.5-VL-7B-Instruct model that has been fine-tuned on olmOCR-mix-1025"; the official HuggingFace card (allenai/olmOCR-2-7B-1025) independently confirms it is "built on Qwen2.5-VL-7B-Instruct," 7B params. (2) 7B VLM: confirmed ? Qwen2.5-VL-7B-Instruct is a 7B vision-language model. (3) Fine-tuned THEN RL-trained: confirmed ? both sources describe SFT (one epoch on olmOCR-mix-1025) followed by GRPO RL training with verifiable rewards (RLVR); ordering is correct. (4) VRAM "roughly 16-24 GB class at FP16, less when quantized": neither paper nor card states inference VRAM explicitly, but the claim is a hedged engineering inference ("roughly," "class") that is corroborated by multiple independent sources: a 24GB RTX 4090 runs Qwen2.5-VL-7B without issue; FP16 weights are ~16GB and text-7B FP16 inference needs ~17GB (apxml.com, Novita AI); quantization lowers this ? and the official model card explicitly recommends an FP8 quantized version (olmOCR-2-7B-1025-FP8), directly supporting "less when quantized." Minor nuance: BF16 weights are ~8B-param file size per the card, but this reinforces rather than contradicts the 16-24GB tier. No contradicting evidence found; claim is current (Oct 2025 release, verified May 2026) and primary-sourced.

[2] On the olmOCR-Bench evaluation, olmOCR 2 scores 84.9 on tables and 83.7 on multi-column layouts, with the paper attributing its

Vote: 3-0 · Source: https://arxiv.org/html/2510.19817v1 (primary)
Quote: "largest improvements in math formula conversion, table parsing, and multi-column layouts"
Verifier evidence (high): CONFIRMED across two independent high-quality sources. PRIMARY (arXiv 2510.19817v1, Table 3): olmOCR 2 final scores are Tables = 84.9, Multi-column = 83.7 (overall 82.4 ± 1.1). The official Ai2 blog (allenai.org/blog/olmocr-2) independently states "84.9% on tables (up from 72.9%), 83.7% on multi-column (up from 77.3%)." The attribution is also verbatim-accurate: the arXiv ABSTRACT reads "...state-of-the-art performance on olmOCR-Bench... with the largest improvements in math formula conversion, table parsing, and multi-column layouts compared to previous versions." So both the numbers (84.9 / 83.7) and the "largest gains attributed to table parsing and multi-column layouts" framing are directly supported. MINOR CAVEATS (do not refute): (a) The source attributes largest gains to THREE areas ? math formula conversion is also listed; the claim drops it (harmless for bank statements, not a misread). (b) "Largest improvements" is a relative DELTA vs. olmOCR v1 (tables +12.0, multi-col +6.4), NOT an absolute ranking ? in absolute terms Base (99.7) and Headers/Footers (96.1) score higher; but the claim correctly says "largest gains," matching the source's "improvements compared to previous versions." (c) olmOCR-Bench is English-only and uses binary unit-test scoring, so generalization to Indian bank statements is unproven ? but that does not contradict the specific verified numbers. No contradicting source found.

[3] Pixtral-12B-2409 model weights are released under the Apache-2.0 license, making the weights commercially safe (not a restrictive

Vote: 3-0 · Source: https://huggingface.co/mistralai/Pixtral-12B-2409 (primary)
Quote: "License: apache-2.0"
Verifier evidence (high): CONFIRMED ? claim stands. Verified against the authoritative primary source: the YAML frontmatter of the official weights repo at https://huggingface.co/mistralai/Pixtral-12B-2409/raw/main/README.md sets exactly `license: apache-2.0`. This is the machine-readable, contractual license declaration on the repo that HOSTS the actual model weights, so it governs the weights (not merely surrounding code). Corroborated by: (a) Mistral's own announcement at mistral.ai/news/pixtral-12b ("License: Apache 2.0"); (b) AWS Bedrock Marketplace blog ("released with open weights under an Apache 2.0 license"); (c) multiple secondary sources noting Pixtral 12B follows the same Apache-2.0 pattern as Mistral 7B / Mixtral / Mistral Nemo. Adversarial check: I specifically searched for a Mistral Research License (MRL)/non-commercial restriction and found NONE applying to the 12B-2409. The only contrary hint was a vague secondary line ("commercial users need a paid license") that the search itself flagged as describing an outdated/different arrangement ? that restriction applies to OTHER Mistral vision models (e.g., Pixtral Large), not to Pixtral-12B-2409. IMPORTANT SCOPING CAVEAT (not a refutation, since the claim is correctly narrow): Apache-2.0 here is specific to Pixtral-12B-2409; do NOT generalize to all Pixtral/Mistral VLMs ? Pixtral Large and similar use the restrictive non-commercial MRL. As written, the claim ("Pixtral-12B-2409 weights are Apache-2.0, commercially safe") is accurate and well-sourced.

[4] Pixtral-12B scores 90.7 ANLS on DocVQA and 81.8 (CoT) on ChartQA, indicating strong document-understanding/chart-reading ca

Vote: 2-1 · Source: https://huggingface.co/mistralai/Pixtral-12B-2409 (primary)
Quote: "DocVQA (ANLS): 90.7 ... ChartQA (CoT): 81.8"
Verifier evidence (high): VERIFIED against primary source + paper. The HuggingFace model card raw README (https://huggingface.co/mistralai/Pixtral-12B-2409/raw/main/README.md) contains the

exact table rows: "| **DocVQA** *(ANLS)* | 90.7 ..." and "| **ChartQA** *(CoT)* | **81.8** ...". The arXiv paper (Table 2, https://arxiv.org/html/2410.07073v2) independently confirms DocVQA (ANLS) 90.7 and ChartQA (CoT) 81.8. Numbers are quoted verbatim; ChartQA is correctly flagged as CoT. The claim's interpretive leap ("indicating strong document-understanding/chart-reading capability relevant to parsing statement tables") is appropriately hedged and directionally sound ? DocVQA and ChartQA ARE the standard document/chart-understanding benchmarks. CAVEATS (qualify, do not refute): (a) scores are self-reported by Mistral via their own re-evaluation pipeline, and the paper itself warns MM eval protocols are non-standardized; (b) DocVQA/ChartQA ? dense multi-column financial-table transcription accuracy, and an independent extraction eval found Pixtral-12B slightly behind Pixtral Large / Gemini 1.5 Flash on recall+precision. But the claim never asserts statement-parsing accuracy ? only "capability relevant to" ? so these caveats do not contradict it. Source quality (primary model card + peer-distributed arXiv paper) matches the claim's modest, hedged strength. Current as of 2025-26.

[5] The InternVL2.5-4B Hugging Face model card states the model is released under the MIT License and incorporates pre-trained Qw

Vote: 3-0 · Source: https://github.com/OpenGVLab/InternVL/issues/900 (primary)
Quote: "In the Hugging Face page for the InternVL2_5-4B model, the license section includes the following statement: "This project is released under the MIT License. This project uses the pre-trained Qwen2.5-3B-Instruct as a component, which is licensed under the Apache License 2.0.""
Verifier evidence (high): The claim is narrowly about what the model card STATES, and is fully verified. Direct WebFetch of https://huggingface.co/OpenGVLab/InternVL2_5-4B confirms the License section reads verbatim: "This project is released under the MIT License. This project uses the pre-trained Qwen2.5-3B-Instruct as a component, which is licensed under the Apache License 2.0." The primary source GitHub issue OpenGVLab/InternVL#900 quotes this exact text. The claim's wording ("the card claims is licensed under Apache-2.0") is precise ? it attributes the Apache-2.0 assertion to the card, not to reality.

CRITICAL CAVEAT (must be flagged in the report, do NOT use this claim to assert commercial safety): The card's Apache-2.0 assertion about Qwen2.5-3B-Instruct is FACTUALLY WRONG. The actual Qwen2.5-3B-Instruct weights are under the custom "qwen-research" license, NOT Apache-2.0. WebFetch of https://huggingface.co/Qwen/Qwen2.5-3B-Instruct shows license tag "qwen-research"; the LICENSE file (https://huggingface.co/Qwen/Qwen2.5-3B-Instruct/blob/main/LICENSE) Section 2(a) grants rights "FOR NON-COMMERCIAL PURPOSES ONLY" and 2(b) "If you are commercially using the Materials, you shall request a license from us." Note: among Qwen2.5 sizes, the 3B and 72B are qwen-research (non-commercial); 0.5B/1.5B/7B/14B/32B are Apache-2.0. Issue #900 exists precisely BECAUSE the card's claim is disputed/incorrect. Net effect for muneem: InternVL2.5-4B is NOT commercial-license-safe ? it embeds a non-commercial Qwen2.5-3B component, regardless of InternVL's own MIT release. The verbatim quote is accurate, so refuted=false, but any downstream use of this claim to support commercial safety would be a misread.

[6] The Apache-2.0 attribution in the InternVL2.5-4B model card for its Qwen2.5-3B-Instruct component is disputed as incorrect: the a

Vote: 3-0 · Source: https://github.com/OpenGVLab/InternVL/issues/900 (primary)
Quote: "However, the license for Qwen2.5-3B-Instruct is not Apache 2.0, but rather the Qwen RESEARCH LICENSE."
Verifier evidence (high): CLAIM CONFIRMED via three independent sources; could not be refuted.

1) PRIMARY SOURCE (github.com/OpenGVLab/InternVL/issues/900): Issue titled "About the license of internVL2.5-4B model," opened 2025-02-12 by user NaokiAsatani, status CLOSED. Author cites the InternVL2.5-4B HuggingFace card claiming the embedded Qwen2.5-3B-Instruct is "licensed under the Apache License 2.0," then corrects that it is actually the Qwen RESEARCH LICENSE (which prohibits commercial use), and asks whether InternVL2.5-4B is eligible for commercial use. The supporting quote matches. (Caveat: no visible InternVL-maintainer acknowledgment in the thread ? but this does not affect the underlying license fact.)

2) AUTHORITATIVE WEIGHTS LICENSE (huggingface.co/Qwen/Qwen2.5-3B-Instruct): Model card metadata shows "License: qwen-research" ? NOT Apache 2.0. A HF discussion "Why a proprietary license?" corroborates. This is the definitive source for the weights license.

3) OFFICIAL QWEN BLOG (qwenlm.github.io/blog/qwen2.5/): Exact text ? "All our open-source models, except for the 3B and 72B variants, are licensed under Apache 2.0." This independently confirms the 3B is excluded from Apache-2.0.

ASSESSMENT OF CLAIM WORDING: The claim says the Apache-2.0 attribution "is disputed as incorrect." It is in fact VERIFIABLY incorrect (not merely disputed) ? the InternVL2.5-4B card's Apache-2.0 attribution for its Qwen2.5-3B-Instruct component is factually wrong; the 3B weights carry the non-commercial Qwen Research License. The claim therefore understates the strength of the evidence rather than overreaching. Highly relevant to muneem's commercial-license-safety constraint: InternVL2.5-4B's weights are tainted by the non-commercial 3B base and are NOT commercially safe despite the card's erroneous Apache-2.0 label.

Checklist: (1) supported by quote ? yes; (2) no contradicting evidence, all sources agree; (3) source quality strong (primary issue + official blog + authoritative model card); (4) not outdated (Feb 2025 issue; qwen-research tag current); (5) not marketing/speculation ? verifiable license-file fact.

[7] The Qwen Research License prohibits commercial use, which casts doubt on whether the InternVL2.5-4B weights can legally be us

Vote: 3-0 · Source: https://github.com/OpenGVLab/InternVL/issues/900 (primary)
Quote: "Since the Qwen RESEARCH LICENSE prohibits commercial use, I would like to confirm whether the InternVL2_5-4B model is eligible for commercial use under these circumstances."
Verifier evidence (high): CLAIM IS CONFIRMED, not refuted. Verified against primary sources (not just the cited forum issue):

1. BASE MODEL CORRECT: The official InternVL2.5-4B model card (huggingface.co/OpenGVLab/InternVL2_5-4B) lists "Qwen2.5-3B-Instruct" as the Language Part for InternVL2_5-4B.

2. QWEN RESEARCH LICENSE DOES PROHIBIT COMMERCIAL USE: The actual LICENSE file at huggingface.co/Qwen/Qwen2.5-3B-Instruct/blob/main/LICENSE is the "Qwen RESEARCH LICENSE AGREEMENT," Section 2.a: grants rights "FOR NON-COMMERCIAL PURPOSES ONLY"; Section 2.b: "If you are commercially using the Materials, you shall request a license from us." The model card's license metadata field reads `qwen-research`.

3. THE 3B SIZE IS SPECIFICALLY THE EXCEPTION: In the Qwen2.5 family, only the 3B and 72B sizes use the Qwen Research License; 0.5B/1.5B/7B/14B/32B are Apache-2.0. InternVL2.5-4B is built on the 3B, so it lands precisely on a non-commercial base.

4. INTERNVL'S OWN ATTRIBUTION IS WRONG: InternVL2.5-4B's model card claims its Qwen2.5-3B-Instruct component "is licensed under the Apache License 2.0" ? directly contradicting Qwen's actual license file. This is exactly the discrepancy issue #900 flagged.

5. LEGAL DOUBT IS REAL: A derivative work (MIT) cannot grant rights to upstream weights it doesn't own; the non-commercial restriction on the Qwen base propagates to the merged weights regardless of InternVL's MIT label. Issue #900 was closed with NO maintainer response resolving the question, so the doubt remains genuinely unresolved.

The claim's hedged framing ("casts doubt") is appropriately calibrated ? it does not overclaim a definitive prohibition. Source quality is strong: the substantive facts are confirmed by the authoritative HuggingFace license files themselves, with the forum issue as corroboration. Current as of the live model card (license field = qwen-research). No contradicting evidence found; multiple independent sources confirm the 3B/72B non-Apache exception. For muneem's commercial-license-safety requirement, InternVL2.5-4B (and the 2B variant, also Qwen2.5-based) should be FLAGGED as commercially unsafe; InternVL2.5 variants built on InternLM2.5 (e.g., 8B/26B, which use Apache-2.0 internlm-chat) would be the license-safe alternative within the family.

[8] Surya's model WEIGHTS are NOT permissively licensed: they use a modified AI Pubs OpenRAIL-M license that is free only for rese

Vote: 3-0 · Source: https://github.com/VikParuchuri/surya/blob/master/LICENSE (primary)
Quote: "The model weights use a modified AI Pubs Open Rail-M license (free for research, personal use, and startups under $5M funding/revenue)"
Verifier evidence (high): VERIFIED via primary source. The raw MODEL_LICENSE file at https://raw.githubusercontent.com/datalab-to/surya/master/MODEL_LICENSE (the repo moved from VikParuchuri/surya to datalab-to/surya, same terms persist on active master) is titled "AI PUBS OpenRAIL-M License" v0.1 and contains verbatim: "(a) for any purpose if You (your employer, or the entity you are affiliated with) generated more than two million US Dollars ($5,000,000) in gross revenue in the prior year, except where Your Use is limited to personal use or research purposes; (b) ... has raised more than two million US dollars ($5,000,000) in total equity or debt funding ..., except where Your Use is limited to personal use or research purposes." The README badge confirms "Code License Apache-2.0 / Model License OpenRAIL-M." PyPI (surya-ocr) independently states the weights use "a modified AI Pubs Open Rail-M license (free for research, personal use, and startups under $5M funding/revenue)" with broader commercial use requiring a

paid license. So: code is Apache-2.0 but the WEIGHTS are NOT permissive ? free only for research/personal/under-threshold orgs, commercial license required above it. This correctly fails muneem's requirement that model weights be MIT/BSD/Apache-2.0 for unrestricted commercial use. Two minor non-fatal caveats: (1) the claim's cited URL (.../blob/master/LICENSE) actually points to the Apache *code* license; the OpenRAIL-M weights terms live in the sibling MODEL_LICENSE file ? citation path is imprecise but substance is correct and in the same repo. (2) The license text itself has an internal drafting inconsistency (word form "two million" vs numeral "$5,000,000"); $5M is the numerically stated and universally reported figure. Neither caveat undermines the core claim.

[9] Surya's CODE (the repository LICENSE file at master) is now Apache License 2.0, a permissive license with no revenue restriction

Vote: 3-0 · Source: https://github.com/VikParuchuri/surya/blob/master/LICENSE (primary)
Quote: "Apache License
Version 2.0, January 2004
http://www.apache.org/licenses/"
Verifier evidence (high): VERIFIED, claim survives. Two independent fetches of the LICENSE file at master (GitHub blob https://github.com/VikParuchuri/surya/blob/master/LICENSE AND raw https://raw.githubusercontent.com/VikParuchuri/surya/master/LICENSE) both confirm the file's top reads "Apache License Version 2.0, January 2004" with standard Apache 2.0 terms and NO revenue caps or commercial-use limitations ? the supporting quote is accurate, not cherry-picked. The "stale GPL-3.0" framing is independently confirmed: Surya was first released Jan 2024 under GPL-3.0, then the CODE was relicensed to Apache 2.0, so the "GPL-3.0 code" label is indeed outdated for current master. The weights-only restriction is confirmed by the current official repo README (datalab-to/surya): "The Surya code is licensed under Apache 2.0" while weights use "a modified AI Pubs Open Rail-M license (free for research, personal use, and startups under $5M funding/revenue)." This directly supports all three parts of the claim: (a) code LICENSE at master = Apache 2.0, (b) "GPL-3.0 code" framing is stale, (c) only the weights carry the restrictive license. Source is PRIMARY (the actual LICENSE file), matching the claim's strength. NUANCE (non-refuting): the weights license is precisely "modified AI Pubs Open RAIL-M," and there historically was a GPL dimension for inference ("remove the GPL license requirements for inference" per dual-licensing contact), but the claim is narrowly about the CODE repo license being Apache 2.0, which is correct and well-supported. No contradicting source found.

[10] olmOCR-2-7B-1025 model weights are licensed Apache-2.0 (commercially permissive), clearing muneem's hard license constrain

Vote: 3-0 · Source: https://huggingface.co/allenai/olmOCR-2-7B-1025 (primary)
Quote: "This model is licensed under Apache 2.0. It is intended for research and educational use in accordance with Ai2's Responsible Use Guidelines."
Verifier evidence (high): VERIFIED, claim stands. Primary source confirmation: the raw README.md YAML frontmatter at https://huggingface.co/allenai/olmOCR-2-7B-1025/raw/main/README.md contains `license: apache-2.0`, and the model-card body states verbatim: "This model is licensed under Apache 2.0. It is intended for research and educational use in accordance with Ai2's Responsible Use Guidelines." This matches the supporting quote exactly.

Refutation attempts all FAILED:
(1) Base-model taint check: base is Qwen/Qwen2.5-VL-7B-Instruct, which is itself licensed apache-2.0 (the 7B-Instruct size is Apache-2.0; note only larger Qwen2.5-VL sizes like 72B carry the restrictive Qwen community license). No taint.
(2) Training-data taint check: dataset olmOCR-mix-1025 is ODC-BY (Open Data Commons Attribution ? permissive, attribution-only, not non-commercial).
(3) "Non-binding overlay" characterization is well-supported: the Apache-2.0 grant is the binding instrument; "intended for research and educational use" is aspirational/intent language, not a license restriction clause. Multiple independent sources (HF OLMo-2 cards, Ai2 positioning) confirm Ai2 explicitly permits and encourages COMMERCIAL use of its Apache-2.0 OLMo-family weights, treating the Responsible Use Guidelines as advisory ethics, not a legal restriction.

Minor caveat (does NOT refute): Ai2's Responsible Use Guidelines page does say the Guidelines "govern your use" of Ai2 artifacts, so a maximally conservative legal reading might treat the overlay as more than purely cosmetic. But this is exactly the nuance the claim already honestly flags ("non-binding ... Responsible Use Guidelines overlay"), so the claim is not an overreach. Claim is current (1025 = Oct 2025 release), primary-sourced, and not marketing fluff.

[11] olmOCR-2-7B is a document-OCR-specialist VLM fine-tuned from Qwen2.5-VL-7B-Instruct, with additional GRPO RL training spec

Vote: 3-0 · Source: https://huggingface.co/allenai/olmOCR-2-7B-1025 (primary)

Quote: "This is a release of the olmOCR model that's fine tuned from Qwen2.5-VL-7B-Instruct using the olmOCR-mix-1025 dataset. It has been additionally fine tuned using GRPO RL training to boost its performance at math equations, tables, and other tricky OCR cases."
Verifier evidence (high): Claim VERIFIED against primary source and two corroborating sources; could not be refuted.

PRIMARY SOURCE (official Ai2 model card, https://huggingface.co/allenai/olmOCR-2-7B-1025) states verbatim: "This is a release of the olmOCR model that's fine tuned from Qwen2.5-VL-7B-Instruct using the olmOCR-mix-1025 dataset. It has been additionally fine tuned using GRPO RL training to boost its performance at math equations, tables, and other tricky OCR cases." This matches the claim almost word-for-word.

CORROBORATION 1 (official Ai2 blog, https://allenai.org/blog/olmocr-2): confirms olmOCR 2 "was built on Qwen2.5-VL-7B," fine-tuned on olmOCR-mix-1025 (270k PDF pages), and used "Group Relative Policy Optimization (GRPO)" with unit-test rewards covering "table structure, equation accuracy, reading order, etc." Backed by paper arXiv 2510.19817 ("olmOCR 2: Unit Test Rewards for Document OCR").

CORROBORATION 2 (FP8 variant card, https://huggingface.co/allenai/olmOCR-2-7B-1025-FP8): identical fine-tuning/GRPO description; license "Apache 2.0."

CHECKLIST: (1) Quote supports claim ? yes, near-verbatim; "tables" is one of the explicitly named GRPO targets, not cherry-picked. (2) No contradicting evidence found across 3 independent sources. (3) Source quality is high ? first-party Ai2 card + blog + technical paper. (4) Not outdated ? "1025" = Oct 2025 release, current as of May 2026 research date. (5) Phrasing originates from Ai2's own card but substance is verified by the technical paper, not marketing-only.

MINOR NUANCE (does not refute): The Ai2 blog clarifies GRPO's mechanism rewards span table structure, equation accuracy AND reading order via unit tests ? i.e. not exclusively math/tables. But tables ARE explicitly named as a GRPO improvement target, so the claim's emphasis is accurate, not an overreach. The claimant's addendum "the harder cases for bank-statement parsing" is interpretation outside the cited quote but is reasonable. The "document-OCR-specialist VLM" descriptor is accurate (olmOCR is purpose-built for document OCR). Apache-2.0 weights license is also confirmed, which is favorable for muneem's commercial-safety constraint.

[12] On olmOCR-Bench the model scores 82.3% overall with 84.3% on Tables and 84.3% on Multi-column layouts ? directly relevant to
Vote: 2-1 · Source: https://huggingface.co/allenai/olmOCR-2-7B-1025 (primary)
Quote: "Tables | 84.3% ... Multi-column | 84.3% ... Overall | 82.3 ± 1.1%"
Verifier evidence (high): VERIFIED against the primary source's raw README HTML (https://huggingface.co/allenai/olmOCR-2-7B-1025/raw/main/README.md). The benchmark table row for "olmOCR pipeline v0.4.0 with olmOCR-2-7B-1025" maps unambiguously by column: ArXiv 82.9 | Old Scans Math 82.1 | Tables 84.3 | Old Scans 48.3 | Headers and Footers 95.7 | Multi column 84.3 | Long tiny text 81.4 | Base 99.7 | Overall 82.3 ± 1.1. All three numbers in the claim (Overall 82.3, Tables 84.3, Multi-column 84.3) are EXACTLY correct for this model, and it is the model named in the claim. The arXiv paper (2510.19817, Oct 2025) backs the same model; current within the May 2026 window.

I attempted to refute via a competing number set surfaced in search (Overall 82.4, Tables 84.9, Multi-column 96.1) ? but those belong to the DIFFERENT olmOCR-2-7B-1025-FP8 variant row (and 96.1 is actually that variant's Headers-and-Footers / shifted-column value, not its multi-column), not a contradiction of the BF16 claim. Initial markdown-converted reads misaligned the columns; the raw HTML resolved it definitively.

ONE caveat (insufficient to refute): the appended gloss "directly relevant to dense, multi-column bank-statement tables" is an inferential framing. olmOCR-Bench's Tables/Multi-column categories cover general documents (papers, scans), NOT Indian bank-statement transaction tables; a high score there does not by itself establish reliability on financial numerics (transposed digits, dropped/duplicated rows). But this is a reasonable topical-relevance statement, not a fabricated empirical claim ? the factual core (the three numbers, attributed to the correct model, from the primary source) is fully supported.

[13] Table Transformer (TATR) repository code is licensed MIT, the pretrained structure-recognition model weights on Hugging Face a
Vote: 3-0 · Source: https://github.com/microsoft/table-transformer (primary)

Quote: "Code: 'MIT license'. Model card (microsoft/table-transformer-structure-recognition): 'License: MIT'. Dataset card (bsmock/pubtables-1m): 'License: cdla-permissive-2.0' (Community Data License Agreement - Permissive 2.0)."
Verifier evidence (high): All three sub-claims verified against primary sources. (1) Repo code = MIT: the official LICENSE file at github.com/microsoft/table-transformer/blob/main/LICENSE contains standard MIT License text ("MIT License"), and the repo footer shows "MIT license". (2) Structure-recognition weights = MIT: HF model card microsoft/table-transformer-structure-recognition metadata header states "License: mit". (3) PubTables-1M dataset = CDLA-Permissive-2.0: HF dataset card bsmock/pubtables-1m metadata states "License: cdla-permissive-2.0", corroborated by independent search (Microsoft Research, cdla.dev) confirming the official dataset is CDLA-Permissive-2.0. Adversarial check on the dataset source: the claim cites a community mirror (bsmock), but bsmock = Brandon Smock, lead author of the PubTables-1M paper and TATR, so it is effectively an author-published mirror, not a mislabeled re-upload; the GitHub README also points to CDLA-Permissive-2.0, so no source conflict. License-safety framing is accurate: CDLA-Permissive-2.0 imposes no restrictions on computational results and is NOT non-commercial, consistent with the claim's contrast against NC doc-AI datasets (e.g. LayoutLMv3 CC-BY-NC). Claim is current (all sources live in the 2025-2026 window) and backed by primary sources matching its strength.

[14] TATR is a DETR-based object-detection model that performs both table detection and table structure recognition (rows, columns,
Vote: 3-0 · Source: https://github.com/microsoft/table-transformer (primary)
Quote: ""A deep learning model based on object detection for extracting tables from PDFs and images." ... "The Table Transformer is a DETR (Detection Transformer)-based object detection model fine-tuned for table structure recognition" ? performing both 'Table Detection' and 'Table Structure Recognition: Understanding cell topology, bounding boxes, and text content within tables.'"
Verifier evidence (high): CLAIM FULLY SUPPORTED by the primary source (Microsoft's official GitHub repo, github.com/microsoft/table-transformer) and independently corroborated.

(1) DETR-based object detection: CONFIRMED. Repo describes it as "a deep learning model based on object detection for extracting tables from PDFs and images," built on the DETR (DEtection TRansformer) architecture (DETR R18 variants). HuggingFace model cards (microsoft/table-transformer-detection, -structure-recognition) confirm the same. Trained on PubTables-1M and FinTabNet.

(2) Performs BOTH table detection AND table structure recognition: CONFIRMED. Two distinct task models ? Table Detection (locating tables in documents) and Table Structure Recognition (rows, columns, cell topology). Matches the quote exactly.

(3) Recognizes cell topology/rows/columns: CONFIRMED. Outputs "full bounding boxes in both image and PDF coordinates for all table rows, columns, and cells (including blank cells)."

(4) Structure-recognition specialist, NOT end-to-end OCR: STRONGLY CONFIRMED ? this is the load-bearing part of the claim and it is explicit in the repo: "TATR is an object detection model that recognizes tables from image input. The inference code built on TATR needs text extraction (from OCR or directly from PDF) as a separate input in order to include text in its HTML or CSV output." Independent search restated: "TATR is not an end-to-end solution but rather requires separate OCR processing."

Source quality matches claim strength (primary official repo + HF model cards). Claim is current (v1.1 models maintained on HF, actively referenced 2025). No contradicting evidence found; sources reinforce the framing. Minor nuance: TATR runs on images and uses PDF coordinates for text mapping, but the official repo itself markets it as working on "PDFs and images," so the claim's wording is not an overreach.

[15] Surya's model WEIGHTS are NOT permissively licensed: they are governed by the 'AI PUBS OPEN RAIL-M LICENSE (MODIFIED)',
Vote: 3-0 · Source: https://github.com/datalab-to/surya/blob/master/MODEL_LICENSE (primary)
Quote: "AI PUBS OPEN RAIL-M LICENSE (MODIFIED)

Version 0.1, March 2, 2023 (Modified)
http://licenses.ai/"
Verifier evidence (high): CLAIM CONFIRMED (cannot refute). Primary source verified: https://github.com/datalab-to/surya/blob/master/MODEL_LICENSE on the current master branch literally opens with "AI PUBS OPEN RAIL-M LICENSE (MODIFIED) / Version 0.1, March 2, 2023 (Modified) / http://licenses.ai/" ? matching the supporting quote verbatim. The file contains

use-based + revenue-threshold restrictions (Section III/Attachment A: cannot use if org had >$5M prior-year gross revenue or raised >$5M equity/debt, except personal/research; no competing products), which propagate to downstream use as enforceable provisions. This is unambiguously NOT a permissive license (not MIT/BSD/Apache/ISC/HPND); it is OpenRAIL-M family with commercial restrictions.

CORROBORATION (independent, not contradictory): Surya's own README on master (https://github.com/datalab-to/surya/blob/master/README.md) states in its "Commercial usage" section: "The model weights use a modified AI Pubs Open Rail-M license (free for research, personal use, and startups under $5M funding/revenue)" ? the maintainer's own characterization confirms the claim. Multiple third-party sources (PyPI surya-ocr, libraries.io, forkable.io) repeat the same OpenRAIL-M-weights / commercial-threshold framing.

CHECKLIST: (1) Quote support ? exact, verbatim from primary source. (2) Contradicting evidence ? none found; all sources agree weights are restricted. (3) Source quality ? primary (license file + official README), highest possible for a licensing claim. (4) Outdated ? no; fetched from live master branch in 2025-2026. (5) Marketing/speculation ? no; it is a legal license document.

NO OVERREACH: The claim is scoped strictly to the WEIGHTS, which is correct. (Note: search results show some confusion about whether the CODE is Apache-2.0 vs GPL ? but the claim under review makes NO assertion about the code license, so that ambiguity does not touch this claim.) The "Version 0.1 / March 2, 2023" date is the license-text version, not a staleness flag ? datalab applies this exact license version to current weights, so the claim correctly attributes the governing license. The conclusion that the weights violate a "permissive weights required" hard constraint (muneem #3) is sound.

[16] Surya's weights forbid commercial use above a revenue threshold: use 'for any purpose' is barred if the user/employer/affiliated e

Vote: 3-0 · Source: https://github.com/datalab-to/surya/blob/master/MODEL_LICENSE (primary)
Quote: "(a) for any purpose if You (your employer, or the entity you are affiliated with) generated more than two million US Dollars ($5,000,000) in gross revenue in the prior year, except where Your Use is limited to personal use or research purposes;"
Verifier evidence (high): VERIFIED against primary source. Fetched the raw license file at https://raw.githubusercontent.com/datalab-to/surya/master/MODEL_LICENSE (current master HEAD). It is the "AI PUBS OPEN RAIL-M LICENSE (MODIFIED)", Version 0.1, March 2, 2023. Attachment A "USE RESTRICTIONS", section 2 "Commercial", subsection (a) reads VERBATIM: "for any purpose if You (your employer, or the entity you are affiliated with) generated more than two million US Dollars ($5,000,000) in gross revenue in the prior year, except where Your Use is limited to personal use or research purposes."

This confirms every element of the claim: (1) commercial use is barred above a revenue threshold; (2) "personal use or research purposes" is explicitly exempted; (3) the source is internally inconsistent ? the spelled-out words say "two million US Dollars" while the parenthetical numeral reads "($5,000,000)". Two independent WebFetch calls of both the raw file and the GitHub blob view reproduced the identical contradictory wording. A WebSearch corroborated the substance (Surya weights "free for research, personal use, and startups under $5M funding/revenue"; commercial license available at datalab.to), and the search summaries themselves disagreed on whether the figure is $2M or $5M ? which reflects (rather than contradicts) the documented internal inconsistency.

The actual license is even MORE restrictive than the claim states (subsection (b) adds a $2M/$5M funding-raised threshold and (c) adds a competitor-use bar), but the claim asserts only the revenue clause + exemption, which is precisely accurate. No overreach, no contradiction, primary source, current. This is a textbook well-supported claim and directly relevant to muneem's commercial-license-safety memo: Surya's WEIGHTS are NOT permissive (not MIT/BSD/Apache) and should be FLAGGED as non-commercial-restricted for the muneem use case (single-user personal use would fall under the personal-use exemption, but the weights cannot be relied on as commercially permissive).

[17] The Surya CODE and the Surya WEIGHTS are licensed separately: the repo's code LICENSE file is Apache-2.0 (permissive), but th

Vote: 3-0 · Source: https://github.com/datalab-to/surya/blob/master/MODEL_LICENSE (primary)
Quote: ""Complementary Material" means the applicable source code and scripts used to define, run, load, benchmark or evaluate the Model, and used to prepare data for training or evaluation, if any."
Verifier evidence (high): PRIMARY SOURCES CONFIRM the claim's core. (1) Surya's repo LICENSE file is Apache-2.0 (verified header: "Apache License Version 2.0, January 2004"). (2) The

separate MODEL_LICENSE is "AI PUBS OPEN RAIL-M LICENSE (MODIFIED), Version 0.1" and explicitly defines "Complementary Material" = "the applicable source code and scripts used to define, run, load, benchmark or evaluate the Model..." and states "Complementary Material is not licensed under this License" ? exactly the carve-out the claim describes. (3) Current README (master) and the HF card datalab-to/surya-ocr-2 (license tag: `openrail`) both state verbatim: "The Surya code is licensed under Apache 2.0. The model weights use a modified AI Pubs Open Rail-M license (free for research, personal use, and startups under $5M funding/revenue)." So the weights ARE the binding constraint and are NON-permissive (use-based restrictions + a hard $5M revenue/funding commercial cap per Attachment A) ? verifying the repo LICENSE alone would indeed wrongly conclude "fully permissive."

ONE NUANCE (does not refute): older descriptions that say "GPL" referred to the CODE (a now-superseded code license ? one search result still describes "the code is GPL" while the current README says Apache-2.0), NOT the weights. The weights have been OpenRAIL-M. The claim's phrasing "NOT the GPL/commercial dual-license that older descriptions cite" conflates the old code license (GPL) with the weights license slightly, but this is immaterial to the verified central point: current code=Apache-2.0, current weights=modified OpenRAIL-M with a $5M commercial restriction. If anything the modified OpenRAIL-M is MORE restrictive than the claim implies, strengthening (not weakening) the research conclusion that Surya weights are commercially unsafe by default and must be FLAGGED.

[18] MinerU is no longer AGPLv3; it is now distributed under the 'MinerU Open Source License', which is the Apache License 2.0 plus
Vote: 3-0 · Source: https://github.com/opendatalab/MinerU/blob/master/LICENSE.md (primary)
Quote: "The license is called the "MinerU Open Source License." It opens with: "MinerU is licensed under Apache License 2.0 and is subject to the additional terms below.""
Verifier evidence (high): CLAIM CONFIRMED via primary source. WebFetch of the actual LICENSE.md at github.com/opendatalab/MinerU/blob/master/LICENSE.md returned verbatim: title = "MinerU Open Source License"; opening line = "MinerU is licensed under Apache License 2.0 and is subject to the additional terms below." It links to apache.org/licenses/LICENSE-2.0 and contains four additional terms (commercial thresholds: >100M monthly active users OR >$20M monthly revenue requires separate licensing; mandatory attribution for online services; automatic termination on breach; Affiliates/Control definitions). No AGPL text remains in the file. Corroborated by PyPI (pypi.org/project/mineru/): "License Expression: LicenseRef-MinerU-Open-Source-License" and "officially moved from AGPLv3 to the MinerU Open Source License, a custom license based on Apache 2.0." The changelog also confirms removal of two AGPLv3 models (doclayout_yolo, mfd_yolov8) and one CC-BY-NC-SA 4.0 model (layoutreader) ? relevant to the weights-license constraint. The claim's wording ("Apache 2.0 plus additional terms") is a precise, non-overreaching match to the primary text. IMPORTANT NUANCE for the report (does not refute this narrow claim): the "additional terms" make this a CUSTOM RESTRICTIVE license, NOT clean Apache-2.0 ? it imposes commercial thresholds + acceptable-use/attribution conditions, which the research question's hard-constraint #3 flags to avoid. The claim is accurate because it explicitly says "plus additional terms" rather than calling it plain Apache-2.0; downstream synthesis must not strip that qualifier. For muneem's single-user personal use the thresholds are irrelevant, so it is practically safe.

[19] A separate commercial license is only required above very high usage thresholds: monthly active users (MAU) exceeding 100 mil
Vote: 3-0 · Source: https://github.com/opendatalab/MinerU/blob/master/LICENSE.md (primary)
Quote: "A separate commercial license is required if you and affiliates reach either threshold: "monthly active users (MAU) exceed 100 million" or "total monthly revenue exceeds USD 20 million.""
Verifier evidence (high): VERIFIED against the primary source (https://raw.githubusercontent.com/opendatalab/MinerU/master/LICENSE.md) and PyPI metadata. The license text states verbatim: "MinerU is licensed under Apache License 2.0 and is subject to the additional terms below," and a separate commercial license is required only if "monthly active users (MAU) exceed 100 million; or total monthly revenue exceeds USD 20 million." Both numbers match the claim exactly, and the revenue threshold is explicitly MONTHLY (USD 20M/month), as the claim states. Below those thresholds, "MinerU may be used for commercial purposes without a separate commercial license," so the underlying Apache-2.0-plus-additional-terms license governs ? matching the claim. The claim correctly hedges by saying "Apache-2.0-BASED" rather than pure Apache-2.0: PyPI declares it as LicenseRef-MinerU-Open-Source-License (a CUSTOM license; v3.1.0, ~April 2026, moved it from AGPLv3 to Apache-2.0-based). This is the current license as of the 2025-2026 window. CAVEATS (do not contradict the claim, which is scoped to "when a commercial license is required"): (a) there is an INDEPENDENT attribution obligation ? anyone providing online services based on MinerU must prominently indicate MinerU is used, regardless of thresholds; (b) violating that attribution duty or failing to obtain the commercial license

triggers AUTOMATIC license termination. For muneem's commercial-safety lens, note this is a custom license with additional terms, NOT vanilla OSI Apache-2.0 ? but the specific threshold claim under review is accurate and primary-sourced. Source quality is sufficient (primary GitHub LICENSE.md + PyPI). The earlier WebSearch summarizer failed to find the thresholds, but the authoritative LICENSE.md and raw text both confirm them.

[20] PP-StructureV3 provides layout detection, table recognition, formula recognition, chart understanding and reading-order restorat

Vote: 2-0 · Source: https://github.com/PaddlePaddle/PaddleOCR/blob/main/docs/version3.x/algorithm/PP-StructureV3/PP-StructureV3.en.md (primary)

Quote: "For Python projects, you can directly integrate using the PaddleOCR Python API"

Verifier evidence (high): Claim fully confirmed by three independent fetches. (1) PRIMARY SOURCE (GitHub PP-StructureV3.en.md) lists all six claimed capabilities ? layout detection, table recognition, formula recognition, chart understanding, reading-order restoration (plus seal recognition, not claimed) ? states "For Python projects, you can directly integrate using the PaddleOCR Python API," and confirms "the ability to convert results into Markdown files." (2) WebSearch corroborates verbatim: PP-StructureV3 "strengthened the ability of layout detection, table recognition, and formula recognition... added the ability to understand charts and restore reading order, as well as the ability to convert results into Markdown files," with Python usage `from paddleocr import PPStructureV3` and outputs in Markdown/JSON/HTML. (3) OFFICIAL DOCS SITE (paddleocr.ai, main/version3.x) gives the concrete Python code `from paddleocr import PPStructureV3; pipeline = PPStructureV3(); output = pipeline.predict(...)` and the Markdown export method `res.save_to_markdown(save_path="output")`, plus `concatenate_markdown_pages` for multi-page PDFs. The supporting quote directly backs the Python-API portion and is NOT an overreach ? every sub-claim (each capability, Python integration, Markdown export) is independently verified. Source quality is primary/official; documentation is current v3.x (consistent with v3.3.0 docs), not outdated. These are concrete, verifiable engineering features with method names, not cherry-picked benchmarks or marketing fluff. No contradicting evidence found. NOTE: claim is purely about FEATURES/Python-usability ? it makes no assertion about license safety or accuracy on dense Indian bank-statement tables, so this verification does not vouch for those (separate concerns for the research question).

Refuted claims (for transparency)

- "PP-StructureV3's lightweight configuration runs with roughly 7.4 GB peak VRAM (7456 MB) at ~0.61s average latency per page on an A100, making the full layout+table pipeline feasible on consumer 8-12 GB NVIDIA GPUs." (https://github.com/PaddlePaddle/PaddleOCR/blob/main/docs/version3.x/algorithm/PP-StructureV3/PP-StructureV3.en.md, vote 1-0)
- "On table extraction accuracy (table edit-distance, lower is better), the PP-StructureV3 OCR+table-structure pipeline outperforms large general-purpose VLMs and pipeline competitors: it scores 0.159 (EN) / 0.109 (ZH) on Table Edit, versus Qwen2.5-VL-72B at 0.341/0.262, GOT-OCR at 0.459/0.52, Marker-1.2.3 at 0.619/0.685, MinerU-0.9.3 at 0.18/0.344, and Mathpix at 0.243/0.32. This supports a deterministic-pipeline-over-VLM recommendation for dense tables." (https://github.com/PaddlePaddle/PaddleOCR/blob/main/docs/version3.x/algorithm/PP-StructureV3/PP-StructureV3.en.md, vote 0-0)
- "OmniDocBench is a CVPR 2025 document-parsing benchmark by OpenDataLab (Shanghai AI Lab) covering 1,651 PDF pages across 10 document types ? and it explicitly includes financial reports as one of the document categories, making its table-recognition results directly relevant to bank-statement parsing." (https://github.com/opendatalab/OmniDocBench, vote 0-0)
- "The benchmark evaluates table recognition with a dedicated Table TEDS metric, providing an objective, comparable accuracy measure across both OCR/pipeline tools and VLMs specifically for table-structure reconstruction ? exactly the capability muneem needs for dense, semi-ruled statement tables." (https://github.com/opendatalab/OmniDocBench, vote 0-0)

Instructions

1. Identify claims that say the same thing ? merge them, combine their sources.
2. Group related claims into coherent findings. Each finding should directly address the research question.
3. Assign confidence per finding: high (multiple primary sources, unanimous votes), medium (secondary sources or split votes), low (single source or blog-quality).
4. Write a 3-5 sentence executive summary answering the research question.
5. Note caveats: what's uncertain, what sources were weak, what time-sensitivity applies.
6. List 2-4 open questions that emerged but weren't answered.

Structured output only.

2. assistant (2026-05-31T04:26:38.383Z)

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